

8/27/26, 7:08 PM

Batch Tolerance Settings

Configure per-batch quality limits:

Max defect length (mm)

2.00

-

+

Max defect area (mm²)

5.00

-

+

Pixel size (mm/pixel)

0.10

-

+

Active Limits:

Max Length: 2.0 mm

Max Area: 5.0 mm²

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ManuVision AI

AI-Powered Manufacturing Defect Detection & Analysis

Upload Product Image

012.png479.2KB

+

Inspection Result

DEFECT DETECTED: COLOR

Confidence100.0%

Defect Type: color

Uploaded Image

Defect Segmentation & Measurement

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ManuVision AI

Recommended disposition: Rework (The affected nuts can be salvaged through proper re-coating processes.)

Process Parameters to Check

Coating Application Pressure: Verify that the spray pressure is set between 2.5 to 3.5 bar to ensure proper atomization and coverage.

Surface Preparation Temperature: Ensure that the surface cleaning process is conducted at a temperature of 60-80°C to effectively remove contaminants.

Curing Temperature: Check that the curing temperature for the coating is maintained at 150-180°C for the specified duration to ensure proper adhesion and color consistency.

Feed Rate during Coating: Monitor the feed rate during the coating application, ideally set between 200-300 mm/min, to ensure consistent application speed and coverage.

Inspection Dashboard

Total Inspec...Good PartsDefectiveRejected (0...

140146

Defect Type Breakdown:

color: 13 occurrences

scratch: 1 occurrences

Recent Inspections

DEFECT | color | 100.0% conf | Over Tol | L:9.4mm A:46.46mm² | 2026-08-27 13:37:32

Defect Measurements vs Tolerance

Defect Region 1: Length: 9.4mm | Area: 46.46mm² → EXCEEDS TOLERANCE

Length 9.4mm exceeds limit 2.0mm

Area 46.46mm² exceeds limit 5.0mm²

VERDICT: REJECT — Defect exceeds batch tolerance limits

AI Root Cause Analysis

Root Cause Analysis

Most Probable Cause

The most likely cause of the color defect observed on the metal nut is improper surface treatment or finishing processes. This could be due to inadequate cleaning prior to coating, incorrect application of surface treatments (such as anodizing or plating), or contamination of the coating materials. If the surface was not properly prepared, it could lead to uneven color distribution or discoloration after the treatment process.

Contributing Factors

1. Inconsistent Coating Application: Variability in the spray pattern or application technique during the

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DEFECT | color | 98.7% conf | Within Tol | 2026-08-26 15:17:49

DEFECT | color | 98.7% conf | Within Tol | L:1.7mm A:3.5mm² | 2026-08-26 13:40:38

DEFECT | color | 98.7% conf | Over Tol | L:22.5mm A:288.63mm² | 2026-08-26 13:40:06

DEFECT | color | 98.7% conf | Within Tol | 2026-08-26 13:39:33

DEFECT | color | 98.7% conf | Within Tol | 2026-08-26 13:39:08

DEFECT | color | 98.7% conf | Over Tol | L:3.3mm A:7.28999999999977mm² | 2026-08-26 13:36:31

DEFECT | color | 98.7% conf | Within Tol | 2026-08-26 13:36:04

DEFECT | color | 98.7% conf | Over Tol | L:22.6mm A:300.2799999999996mm² | 2026-08-26 13:34:52

DEFECT | color | 98.7% conf | Within Tol | 2026-08-26 13:29:22

coating process could lead to uneven color coverage on the nuts.

2. **Material Contamination:** The presence of oils, dust, or other contaminants on the surface of the nuts before coating could affect the adhesion and appearance of the finish.

3. **Improper Temperature Control:** If the curing or drying temperature of the coating was not maintained within specified limits, it could cause color variations or defects in the finish.

Recommended Corrective Actions

1. Immediate Action: Isolate the affected batch of nuts and conduct a visual inspection to assess the extent of the color defect. If the defect is isolated, consider reworking the affected nuts by stripping the coating and reapplying it after proper surface preparation.

2. Process Adjustment: Review and adjust the coating application parameters, including spray pressure and distance from the surface, to ensure uniform coverage. Implement stricter controls on the cleaning process prior to coating to eliminate contaminants.

3. Preventive Measure: Establish a routine audit of the surface treatment process, including checks on the cleanliness of the nuts before coating and regular calibration of the coating equipment to ensure consistent application.

Quality Impact Assessment

Severity: Medium (The color defect may affect aesthetic quality and customer satisfaction, but does not compromise structural integrity.)

Likelihood of recurrence: Medium (If the root cause is not addressed, similar defects may occur in future batches.)

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